

Cumulative frequency tables and histograms

1 Consider the following table:

- a Find the total number of scores recorded.
- b Find the number of times a score of 14 occurred.
- c Find the number of times a score less than 13 occurred.

Score (x)	Cumulative frequency (cf)
10	7
11	15
12	18
13	20
14	26

2 Complete the following table:

Score (x)	Frequency (f)	Cumulative frequency (cf)
18	10	
19	9	
20	3	22
21		
Total:	29	

3 The number of sightings of the Northern Lights were recorded across various Canadian locations over a period of 1 month. The numbers below represent the number of sightings at each location:

12, 8, 9, 8, 11, 7, 7, 11, 10, 9, 9, 11, 7, 10, 11, 7, 8, 9, 11, 9

- a Construct a cumulative frequency table for this data.
- b Find the number of locations where there were at least 11 sightings.
- c Find the number of locations where there were less than 11 sightings.
- d Find the median number of sightings across all 20 locations.

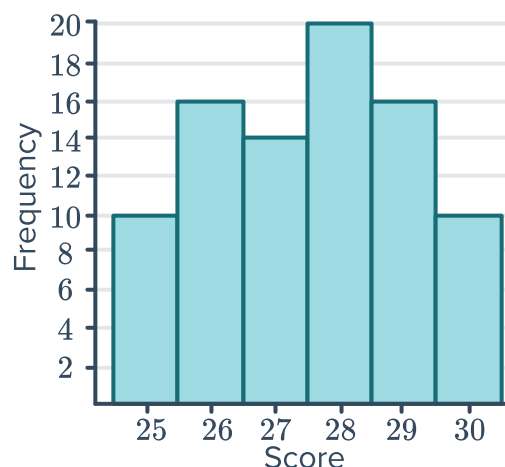
- 4 A pair of dice are rolled 50 times and the numbers appearing on the uppermost face are added to give a score. The results are recorded in the given table:



Score (x)	Frequency (f)	Cumulative frequency (cf)
2	1	
3	2	
4	5	
5	5	
6	5	
7	9	
8	7	
9	5	
10	8	
11	1	
12	2	

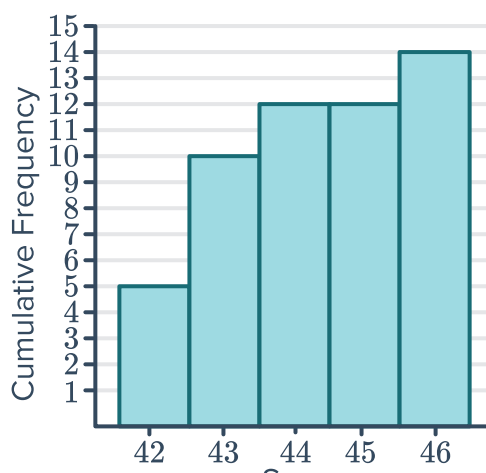
- State the lowest possible score when a single pair of dice are rolled.
- State the highest possible score when a single pair of dice are rolled.
- Complete the table by finding the cumulative frequency values.
- Find the number of times a score of 8 occurred.
- Find the number of times a score more than 9 occurred.
- Find the number of times a score of at most 6 occurred.

- 5 Construct a cumulative frequency table for the data represented in the given histogram:



- 6 Consider the following cumulative frequency histogram:

- Find the total number of scores recorded.
- Find the number of times a score of 46 occurred.
- Find the number of times a score of 45 occurred.
- Find the percentage of scores that were 43 or less. Round your answer to one decimal place.



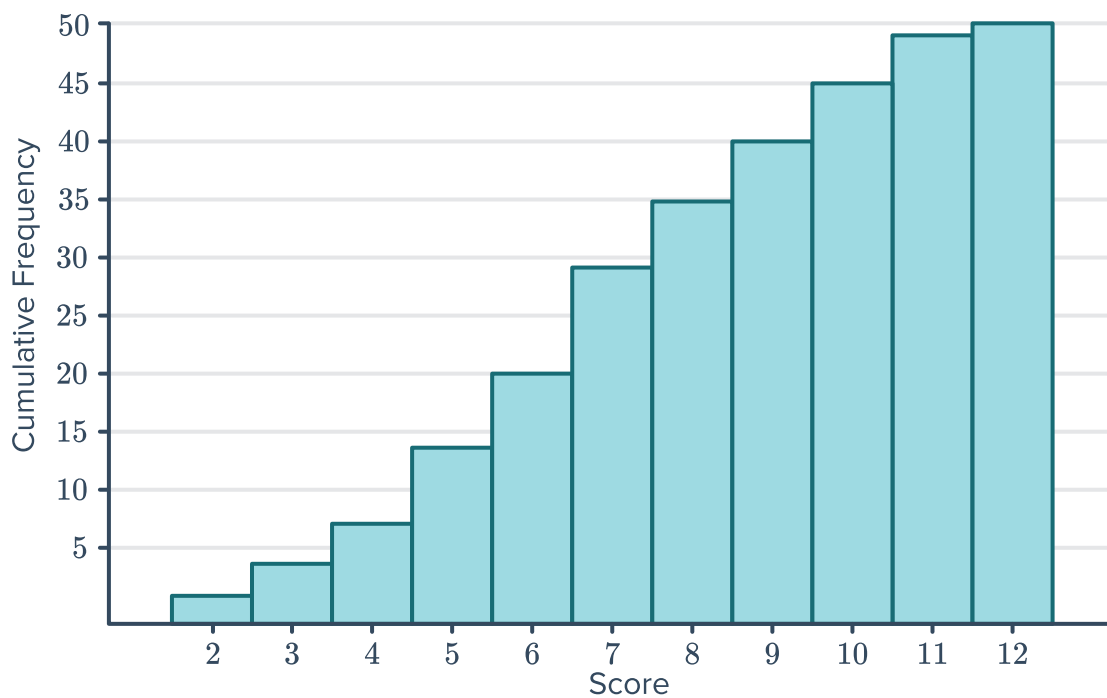


7 Consider the frequency table showing the number of 'holes in one' across golf tournaments:

Number of 'holes in one'	Tournaments
2	5
3	1
4	3
5	4
6	0

- a Construct cumulative frequency histogram for this data.
- b Find the total number of 'holes in one' across all the tournaments.
- c In how many tournaments were at most 3 'holes in one' scored?

8 A pair of dice are rolled 50 times and the numbers appearing on the uppermost face are added to give a score. The results are presented in the following histogram:



- a Explain how the frequency of a certain score can be determined from a cumulative frequency histogram.
- b What was the most frequent score?

- 9 The marks received by a group of 20 students are given below:



75, 75, 95, 60, 95, 75, 60, 75, 95, 70, 70, 70, 60, 83, 75, 83, 70, 60, 75, 70

- a Construct a cumulative frequency histogram for this data.
 - b Find the number of students who scored above 70.
 - c Find the number of students who scored less than or equal to 70.
 - d Find the number of students who scored above 80 but less than or equal to 90.
 - e Find the mode.
 - f Find the mean of the scores, correct to two decimal places.
 - g Find the median of the scores.
- 10 Families were asked how many times they got the flu during winter. The information has been partially filled out in the following table:

- a Complete the frequency and cumulative frequency values in the given table.
- b Find the total number of families that responded.
- c Find the median number of times a family got the flu.
- d Find the mean number of times a family got the flu. Round your answer to two decimal places.
- e Construct a cumulative frequency histogram for the data.
- f Find the number of families who got the flu more than twice.

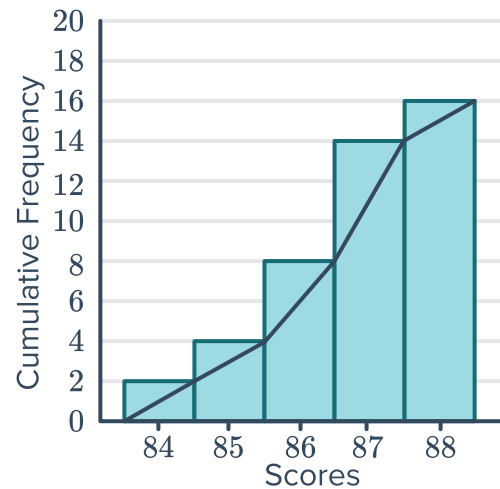
Score (x)	Frequency (f)	fx	Cumulative frequency
0	0	0	
1		2	
2		6	
3		6	
4		8	



Cumulative frequency ogive

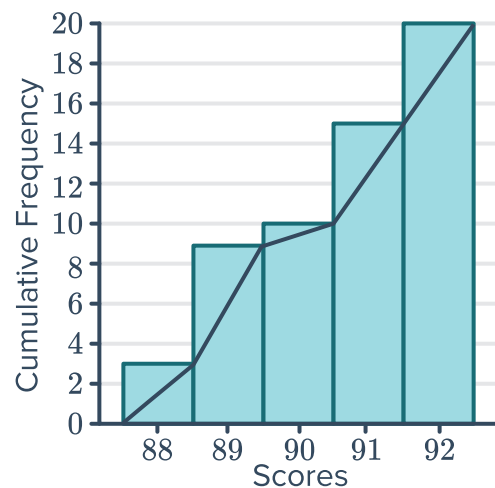
- 11 Use the cumulative frequency histogram to complete the frequency table:

Score	Frequency
84	
85	
86	
87	
88	



- 12 Consider the following cumulative frequency histogram and ogive:

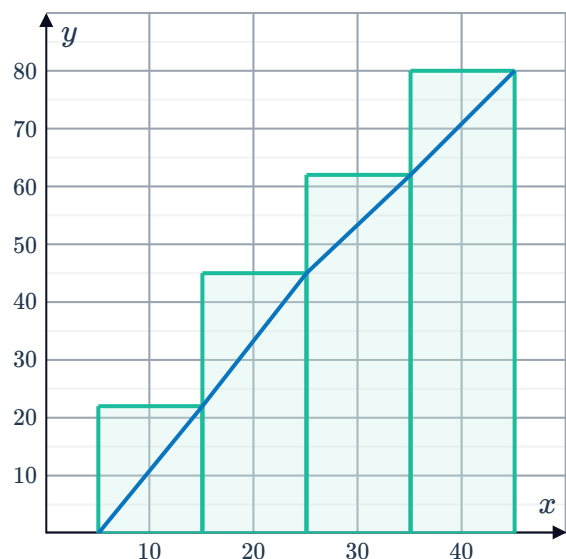
- a Use the ogive to determine the median score.
- b Find the 14th lowest score.
- c How many scores of 92 or less were there?
- d How many scores fewer than 90 were there?



- 13 Consider the following cumulative frequency histogram and ogive:

Estimate:

- a The 60th percentile.
- b The 10th percentile.
- c The lower quartile.
- d The median
- e The upper quartile.
- f The interquartile range.

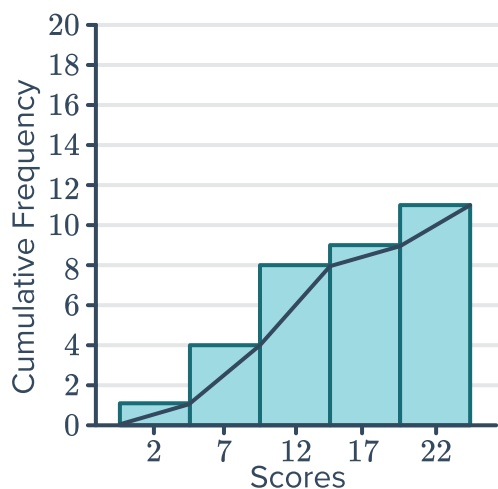


14 For each of the following graphs use the ogive to estimate:

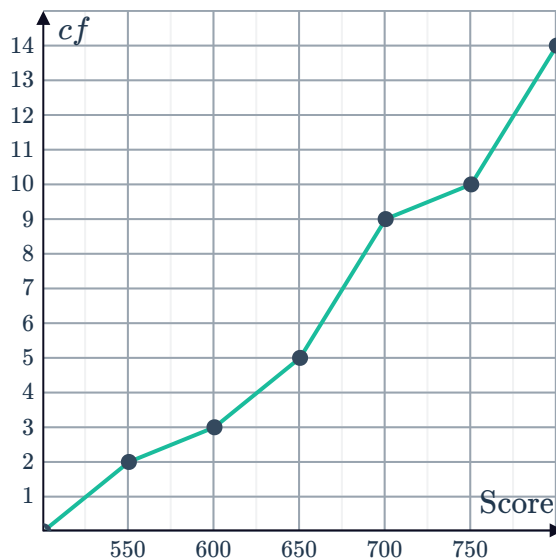


- i The median score.
- ii The lower quartile.
- iii The upper quartile.

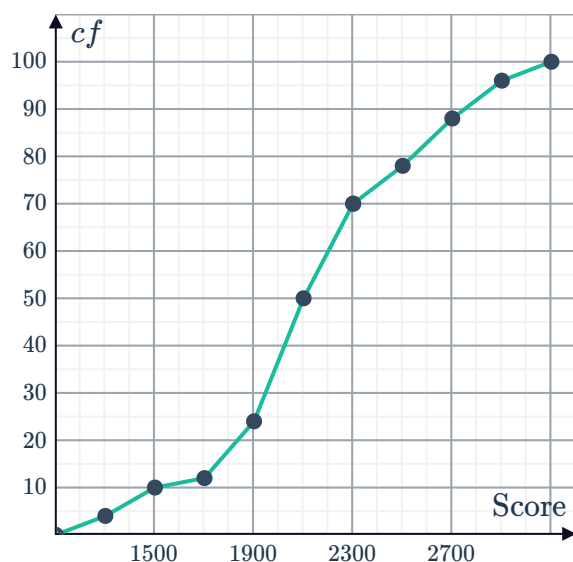
a



b



c



15 Consider the following frequency table:

- a Complete the table by finding the cumulative frequency values.
- b Construct a cumulative frequency histogram and ogive for the data.
- c Find the median.
- d Use your graph to estimate the 20th percentile.

Score (x)	Frequency (f)	Cumulative frequency (cf)
134	3	
135	2	
136	4	
137	6	
138	4	
Total:	19	



Cumulative frequency and grouped data

16 For each of the frequency distribution tables below:

- i Complete the table.
- ii State the modal class.
- iii In which class interval does the median lie?
- iv Using the class centres, estimate the mean to two decimal places.

a

Class	Class centre	Frequency	Cumulative frequency
1 – 7		11	
8 – 14		14	
15 – 21		10	
22 – 28		15	
29 – 35		24	
Total:			

b

Class	Class centre	Frequency	Cumulative frequency
1 – 9		8	
10 – 18		16	
19 – 27		4	
28 – 36		21	
37 – 45		16	
Total:			

17 Consider the frequency distribution table below:



Class	Class centre	Frequency	Cumulative frequency
1 – 9		3	
10 – 18		4	
19 – 27		3	
28 – 36		3	
37 – 45		8	
Total:			

- Complete the table.
- Construct a cumulative frequency histogram for the data.
- State the modal class.
- In which class interval does the median lie?

18 Consider the frequency distribution table below:

Score (x)	Frequency (f)	Cumulative frequency (cf)
1 – 4	4	
5 – 8	7	
9 – 12	11	
13 – 16	7	
17 – 20	4	

- Complete the table.
- Calculate the total frequency.
- Identify the class size.
- Describe the shape of the distribution.
- Approximately one third of the scores recorded are greater than what score?

19 Consider the frequency distribution table :

Score (x)	Frequency (f)	Cumulative frequency (cf)
1 – 5	7	
6 – 10	15	
11 – 15	8	
16 – 20	11	
21 – 25	2	
26 – 30	1	

- a Complete the table.
- b Calculate the total frequency.
- c Identify the class size.
- d Describe the shape of the distribution.
- e Approximately half of the scores recorded are greater than what score?

20 Consider the frequency distribution table below:

Score (x)	Frequency (f)	Cumulative frequency (cf)
20 – 24	9	
25 – 29	18	
30 – 34	37	
35 – 39	14	
40 – 44	9	
45 – 49	6	
50 – 54	3	

- a Complete the table.
- b Calculate the total frequency.
- c Identify the class size.
- d Describe the shape of the distribution.
- e Approximately one third of the scores recorded are greater than what score?

21 Consider the set of scores given below:



58, 59, 70, 64, 69, 73, 64, 68, 59, 60, 54, 73

a Complete the following frequency table:

Class interval	Class centre (cc)	Frequency (f)	$f \times cc$	Cumulative frequency
51 – 55				
56 – 60				
61 – 65				
66 – 70				
71 – 75				
Total				

b State the modal class.

c Using the class centres, estimate the mean to one decimal place.

22 A group of high school students wanted to convince their principal that the school needed air-conditioning. They measured the temperature, t , in a classroom at 1 pm every day during February and recorded the results (in $^{\circ}\text{C}$) below:

35, 23, 24, 23, 30, 29, 36, 24, 35, 32, 29, 36, 33, 30, 30,
26, 27, 24, 32, 30, 28, 27, 24, 31, 26, 36, 29, 35, 30

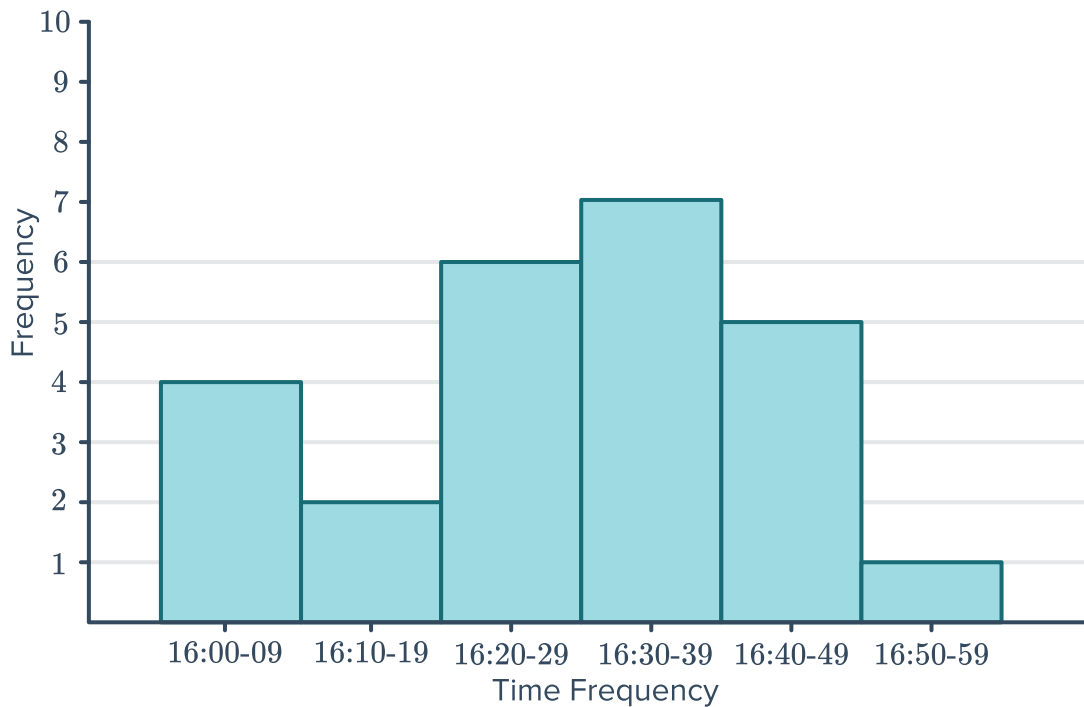
a Construct a cumulative frequency table for the data. Use the discrete intervals of 23 – 25, 26 – 28, and etc.

b Construct a cumulative frequency histogram for the data.

c Find the modal class.

d Using the class centres, estimate the average temperature to two decimal places.

23 A 1500 m swimmer records her time over several training sessions. Her times are recorded in the following histogram:



- Construct a cumulative frequency table for the data using the given intervals.
- Find the total number of training sessions she completed.
- In which class interval does the median time fall?
- Find the number of times she recorded a swim time faster than 16:50.
- Find the percentage of swims that were less than 16:30.

24 The heights of 22 boys in a class are listed below:

133, 150, 133, 142, 149, 146, 135, 150, 143, 140, 151,
151, 132, 149, 149, 138, 142, 149, 136, 147, 143, 145

- Construct a cumulative frequency histogram for the data. Use the discrete intervals of 131 – 135, 136 – 140, etc.
- Find the number of students taller than 150 cm.
- Find the number of students who are at most 150 cm tall.
- Find the number of students taller than 140 cm but shorter than 146 cm.
- State the modal height class of the students.
- Calculate the centre of the class 146 – 150.
- Which class contains the median height?

25 A tree farmer measured the heights of some newly planted saplings to the nearest centimetre. The results were as follows:



20, 25, 25, 25, 24, 30, 25, 29, 24, 15, 24, 15, 34, 25, 30, 25,
14, 14, 35, 19, 25, 15, 15, 29, 35, 20, 25, 30, 25, 30, 24, 25, 24

- a** Find the range.
- b** Construct a cumulative frequency histogram for the data. Use the discrete intervals of 13 – 17, 18 – 22, etc.
- c** Using the class centres, estimate the mean to two decimal places.
- d** In which class interval does the median lie?
- e** State the modal class.
- f** Find the percentage of saplings that are at least 23 cm tall. Round your answer to one decimal place.
- g** Find the percentage of saplings that have a height lower than 23 cm. Round your answer to one decimal place.

- 26** Scientists wanting to determine the effect of fatigue on reaction time while driving. They measured the reaction time, t , of several drivers at night. The results are presented in the following table:



Reaction time (seconds)	Class centre	Frequency	Cumulative frequency
$0.01 \leq t < 0.05$	0.03	29	29
$0.05 \leq t < 0.09$	0.07	32	61
$0.09 \leq t < 0.13$	0.11	39	100
$0.13 \leq t < 0.17$	0.15	31	131
$0.17 \leq t < 0.21$	0.19	36	167
$0.21 \leq t < 0.25$	0.23	33	200

- a** Using the class centres, calculate an estimate of:
- i** The median
 - ii** The mean
- b** One researcher wants to reduce the amount of data by increasing the size of each class interval. Complete the table:

Reaction Time (seconds)	Class Centre	Frequency	Cumulative frequency
$0.01 \leq t < 0.09$			
$0.09 \leq t < 0.17$			
$0.17 \leq t < 0.25$			

- c** Using the new class centres, calculate an estimate of:
- i** The median
 - ii** The mean
- d** By increasing the size of each class interval, by what percentage has the mean changed?